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RESPIRATORY ANOMALIES DETECTION USING CONVOLUTIONAL NEURAL NETWORK

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CONTENT

RESPIRATORY ANOMALIES DETECTION USING CONVOLUTIONAL NEURAL NETWORK

Category	Details
Project Objective	Develop an AI system to detect respiratory diseases based on voice samples.
Process	Data Collection: Collected 2,004 voice samples (Healthy: 1,002, Unhealthy: 1,002). Preprocessing: Extract audio features using MFCC (Mel-frequency cepstral coefficients). Model Development: Deep Learning model using TensorFlow and Keras.

CONTENT

RESPIRATORY ANOMALIES DETECTION USING CONVOLUTIONAL NEURAL NETWORK

Category	Details
Tools and Technologies	Libraries: Python, Librosa (for audio processing), TensorFlow (for deep learning). Format: WAV audio samples for training and prediction.
Outcome	Non-invasive and cost-effective early detection tool for respiratory issues.
Future Scope	Improve model accuracy with more diverse datasets. Real-time application for use in healthcare.

ABSTRACT

This project detects whether a person is healthy or unhealthy by analyzing voice samples. It processes voice features like MFCCs and uses a machine learning model to classify them. The system is non-invasive, quick, and affordable, making it useful for early detection and remote healthcare monitoring. It has the potential to make healthcare more accessible and efficient for everyone.

INTRODUCTION

- Respiratory diseases like asthma, pneumonia, and COPD affect the lungs and breathing, often requiring early diagnosis to prevent severe health complications.
- Deep learning is used to train a neural network to analyze voice patterns and classify them as Healthy or Unhealthy based on respiratory health.
- The dataset was sourced from PhysioNet and ERS websites, and some voice samples were mimicked using AI to enhance the dataset's diversity.
- MFCC (Mel-Frequency Cepstral Coefficients) is a key audio feature extraction technique that converts voice data into numerical representations, making it suitable for machine learning models.

LITERATURE SURVEY

SL NO	Publication Details	Methodology	Research Gap	Link
1	X. Wang, Y. Liu, et al; Lung Disease Recognition Methods Using Audio-Based Analysis with Machine Learning; 2023.	Audio signal-based machine learning models for lung disease detection.	Focused only on classification; lacks comprehensive evaluation.	Link
2	B. A. Akinyemi, S. A. G. E. Bakare, et al; Predicting Pulmonary Function from the Analysis of Voice: A Machine Learning Approach; 2022.	Machine learning methods were applied to voice signals to predict pulmonary function.	Limited dataset size and lack of real-time applicability.	Link

LITERATURE SURVEY

SL NO	Publication Details	Methodology	Research Gap	Link
3	Mobile phone app accurately detects COVID-19 infection in people's voices with AI; ERS News and Features; 2022.	AI-based app for detecting COVID-19 infection from voice data..	Limited to COVID-19 detection, not general respiratory issues.	Link
4	N. B. Thakur, J. Singh, et al; Respiratory Diseases Diagnosis Using Audio Analysis and Artificial Intelligence: A Systematic Review; 2021.	A systematic review of AI techniques used in respiratory diagnosis.	No implementation of advanced deep learning methods.	Link

SYSTEM ARCHITECTURE/ FLOW DIAGRAM

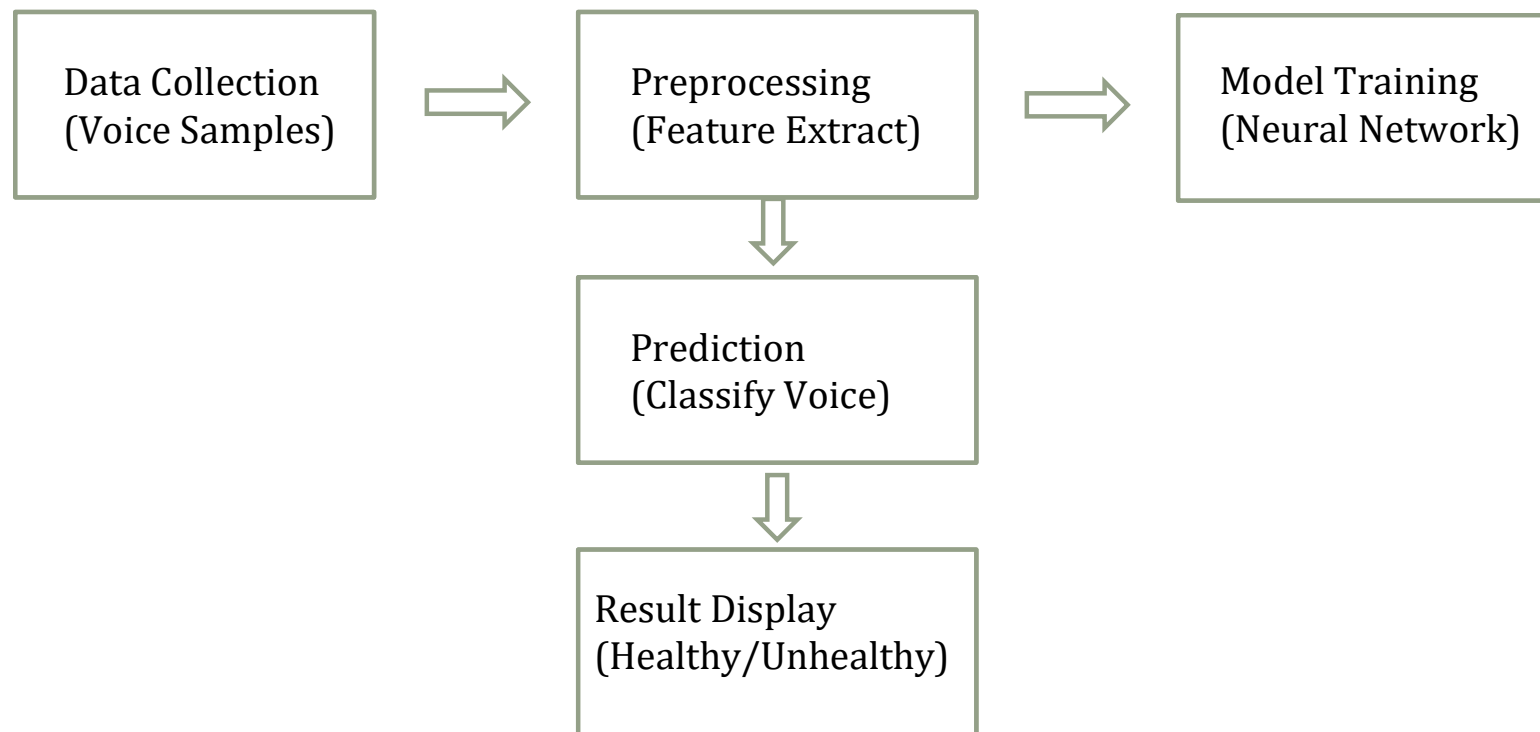


Fig: Respiratory Anomalies Detection

METHODOLOGY

1. Data Collection

- **Voice Samples:** Collected 2,004 voice samples — 1,002 healthy and 1,002 unhealthy samples.
- **Format:** All samples are in .wav format, making them easy to process for feature extraction.

2. Preprocessing

- **Feature Extraction:**
 - Extracted key features like MFCCs (Mel-Frequency Cepstral Coefficients) from each voice sample.
 - Features help to capture the characteristics of voice patterns related to respiratory health.
- **Normalization:** Ensured all data is on the same scale for better performance during model training.

METHODOLOGY

3. Model Building

- **Neural Network:**
 - Used a **simple neural network** model to classify voice samples as **Healthy** or **Unhealthy**.
 - The model learns from the features extracted from the audio samples.

4. Model Training

- **Training:**
 - The model was trained using the **training dataset** (80% of total data).
 - **Optimizer:** Adam optimizer used to improve learning efficiency.

METHODOLOGY

5. Prediction

- **Classification:**
 - When a single voice sample, the model predicts whether it is **Healthy** or **Unhealthy** based on learned features.

6. Model Evaluation

- **Accuracy:** The model's accuracy was tested using the remaining **20% test data**.
- **Evaluation Metrics:** Used accuracy and confusion matrix to see how well the model performs.

CRITICAL ANALYSIS IF ANY

1. Data Limitations

- **Real time Dataset:** Because the model was trained on a limited number of 2,004 samples. A larger dataset or the real time dataset would improve the accuracy and robustness of predictions.
- **Data Imbalance:** The distribution of healthy and unhealthy samples may not represent real-world data, potentially affecting model performance.

2. Feature Extraction Challenges

- **Voice Quality Variations:** Variations in voice quality (e.g., background noise, microphone quality) may affect the feature extraction process, leading to less accurate results.
- **Complexity of Respiratory Sounds:** Not all respiratory issues may be detectable from basic voice features, and certain conditions might not show clear patterns in the voice.

RESULT



MODEL	ACCURACY
Lung Disease Detection Using Audio Signals	80.5 %
Voice-Based Disease Diagnosis Using AI	84.6 %
Machine Learning for Voice Signal Analysis in Medical Diagnosis	87.5 %
Respiratory Anomalies Detection	90 %

Fig: Model Accuracy Comparison

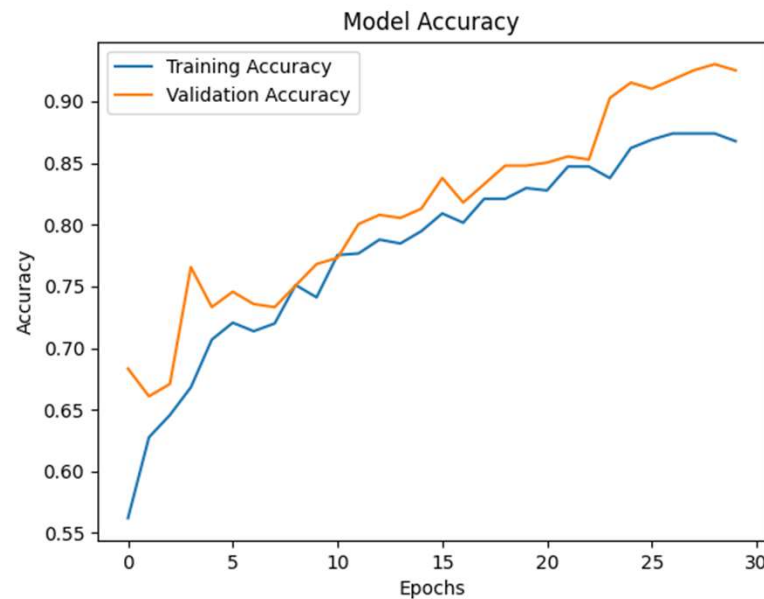
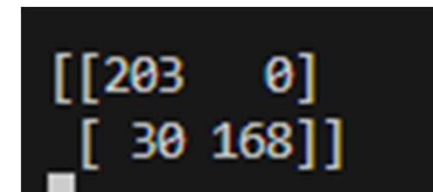


Fig: Model Training and Validation Accuracy

Confusion Matrix:



- True Negatives (TN):** 203 (Healthy samples correctly classified as Healthy).
- False Positives (FP):** 0 (Healthy samples misclassified as Unhealthy).
- False Negatives (FN):** 30 (Unhealthy samples misclassified as Healthy).
- True Positives (TP):** 168 (Unhealthy samples correctly classified as Unhealthy).

Fig: Confusion Matrix

ADVANTAGES

- The system provides a non-invasive and cost-effective method for detecting respiratory problems, making it accessible to a wide range of users. Unlike traditional diagnostic tools, it requires only voice recordings, eliminating the need for expensive tests or physical consultations.
- It is scalable, capable of analyzing a large number of voice samples with minimal infrastructure and cost, making it ideal for widespread adoption.
- **Scalability:** Once deployed, the model can handle a large number of voice samples without additional costs or infrastructure, making it ideal for widespread use.
- **Early Detection:** By identifying respiratory issues at an early stage, the system can help in timely medical intervention, reducing the risk of complications.

FUTURE SCOPE

- The system can be expanded to classify specific respiratory diseases such as asthma, pneumonia, and bronchitis, rather than just differentiating between healthy and unhealthy conditions.
- By including more diverse datasets, the model's accuracy and reliability can be improved significantly, ensuring better performance across various real-world scenarios.
- **Clinical Decision Support:** Integrate the system with hospital systems to assist doctors in making faster and more informed decisions during consultations.
- **Offline Functionality:** Develop lightweight versions of the system that can function offline, especially for regions with limited internet connectivity.
- **Multi-condition Detection:** Train the system to detect co-existing health conditions, such as respiratory problems linked with cardiac issues.

CONCLUSIONS

- This project shows that voice analysis can be used to detect respiratory problems in a simple and non-invasive way. The system successfully classifies voice samples as healthy or unhealthy, providing an affordable and accessible solution for early diagnosis.
- Using voice samples removes the need for expensive tests, making it helpful for people in remote and rural areas. It also gives real-time results, allowing easy and quick health monitoring(In the final project).
- The system can be improved by adding more data, reducing noise, and identifying specific diseases. With these improvements, it can become a powerful tool for better and faster respiratory health care.

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[Link](#)

[2] X. Wang, Y. Liu, et al., *"Lung Disease Recognition Methods Using Audio-Based Analysis with Machine Learning,"* Heliyon, vol. 9, pp. 1-8, 2023.

[Link](#)

[3] N. B. Thakur, J. Singh, et al., *"Respiratory Diseases Diagnosis Using Audio Analysis and Artificial Intelligence: A Systematic Review,"* PLoS Digital Health, vol. 1, no. 3, pp. 1-16, 2021.

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Thank you 😊